

RMO(INDIA) – 1988(1)

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Prove that for every positive integer n , $1^n + 8^n - 3^n - 6^n$ is divisible by 10.
 2. If the letters used in the following expressions denote uniquely a different digit in base 10 and if $(YE)(ME) = (TTT)$, find $E + M + Y + T$.
 3. A printer numbers the pages of a book starting with 1 and uses 3189 digits in all. How many pages does the book have?
 4. For $n > 1$, show that $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ is not an integer.
 5. If $x(y-z)$, $y(z-x)$, $z(x-y)$ are in G.P., find the equation satisfied by the common ratio r .
 6. A person is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return to his cabin. Find the shortest distance (in miles) he has to travel to accomplish this.
 7. Determine all the triangle ABC for which $\tan(A-B) + \tan(B-C) + \tan(C-A) = 0$.
 8. If $\Delta A_1 A_2 A_3$ is equilateral and A_{n+3} is the mid point of the line segment $A_n A_{n+1}$ for all positive integers n , find the measures of $\angle A_{44} A_{45} A_{43}$.
 9. A quadrilateral inscribes a circle and it also circumscribes another circle. If the sides of the quadrilateral are a, b, c, d show that the area of the quadrilateral is $\sqrt{abcd}$.

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